화학컴퓨터 (2009년 1학기)

과제 #3 **(총 135 점)**

1. (**총 10점, 각 5점**)

   (a) 적분인자법 이용

$$\begin{aligned} x\frac{dy}{dx} + 2y = x^3 \quad &\Rightarrow \quad \frac{dy}{dx} + \frac{2y}{x} = x^2 \\ \frac{d}{dx}\left(e^{\int 2/x dx} y\right) = e^{\int 2/x dx} x^2 \quad &\Rightarrow \quad \frac{d}{dx}\left(x^2 y\right) = x^4 \\ x^2 y = \frac{1}{5}x^5 + c \quad &\Rightarrow \quad y = \frac{1}{5}x^3 + \frac{c}{x^2} \end{aligned}$$

   (b) 적분인자법 이용

$$\begin{aligned} \frac{dy}{dx} + \frac{2}{x}y = x^2 + 2 & \\ \frac{d}{dx}\left(e^{\int 2/x dx} y\right) = e^{\int 2/x dx}(x^2+2) \quad &\Rightarrow \quad \frac{d}{dx}\left(x^2 y\right) = x^4 + 2x^2 \\ x^2 y = \frac{1}{5}x^5 + \frac{2}{3}x^3 + c \quad &\Rightarrow \quad y = \frac{1}{5}x^3 + \frac{2}{3}x + \frac{c}{x^2} \end{aligned}$$

2. **(5점)** 적분인자법 이용

$$\begin{aligned} x\frac{dy}{dx} + y = 2x \quad &\Rightarrow \quad \frac{dy}{dx} + \frac{y}{x} = 2 \\ \frac{d}{dx}\left(e^{\int 1/x dx} y\right) = 2e^{\int 1/x dx} \quad &\Rightarrow \quad \frac{d}{dx}\left(xy\right) = 2x \\ xy = x^2 + c \quad &\Rightarrow \quad y = x + \frac{c}{x} \end{aligned}$$

   $y(2) = 2$ 를 윗 식에 대입하면

$$y(2) = 2 \Rightarrow 2 = 2 + \frac{c}{2} \Rightarrow c = 0$$

3. **(총 10점, 각 5점)**

   (a) $y = e^{mx}$ 로 놓으면, 특성식은

$$m^2 - m - 2 = 0 \Rightarrow (m-2)(m+1) = 0 \Rightarrow m = 2 \text{ 또는 } m = -1$$

$$y(x) = c_1 e^{2x} + c_2 e^{-x}$$

   (b) $y = e^{mx}$ 로 놓으면, 특성식은

$$m^2 - 6m + 9 = 0 \Rightarrow (m-3)^2 = 0 \Rightarrow m = 3$$

특성식이 중근을 갖는 경우에 일반해는

$$y(x) = c_1 e^{3x} + c_2 x e^{3x}$$

4. **(총 10점, 각 5점)**

(a) $y = e^{mx}$ 로 놓으면, 특성식은

$$m^2 + 6m = 0 \Rightarrow m(m+6) = 0 \Rightarrow m = -6 \text{ 또는 } m = 0$$

$$y(x) = c_1 e^{-6x} + c_2$$

$y(0) = 0$ 과 $y'(0) = 1$ 을 위 식에 대입하면

$$c_1 + c_2 = 0 \quad \text{그리고} \quad -6c_1 = 1$$
$$c_1 = -\frac{1}{6} \quad \text{그리고} \quad c_2 = \frac{1}{6} \Rightarrow y(x) = \frac{1 - e^{-6x}}{6}$$

(b) $y = e^{mx}$ 로 놓으면, 특성식은

$$m^2 + 4 = 0 \Rightarrow m = \pm 2i$$

특성식이 중근을 갖는 경우에 일반해는

$$y(x) = c_1 e^{2ix} + c_2 e^{-2ix}$$

$y(0) = 0$ 과 $y'(0) = 1$ 을 위 식에 대입하면

$$c_1 + c_2 = 0 \quad \text{그리고} \quad 2ic_1 - 2ic_2 = 1$$
$$c_1 = \frac{1}{4i} \quad \text{그리고} \quad c_2 = -\frac{1}{4i}$$
$$\Rightarrow y(x) = \frac{e^{2ix} - e^{-2ix}}{4i} = \frac{\sin 2x}{2}$$

5. **(총 15점, 각 5점)**

(a)

$$m\frac{dv}{dt} = f - bv \Rightarrow \frac{dv}{dt} + \frac{b}{m}v = \frac{f}{m}. \tag{1}$$

적분인자법을 이용하면,

$$\frac{d}{dt}\left(ve^{(b/m)t}\right) = \frac{f}{m}e^{(b/m)t} \Rightarrow ve^{(b/m)t} = \frac{f}{m}\int e^{(b/m)t}dt$$
$$\Rightarrow ve^{(b/m)t} = \frac{f}{b}e^{(b/m)t} + C, \tag{2}$$

위 식에서 C는 미지 상수이다. where C is a constant. (2) 식으로부터 , $v(t)$는 다음과 같이 쓸 수 있다.

$$v(t) = \frac{f}{b} + Ce^{-(b/m)t}. \tag{3}$$

(b) (3) 식을 이용하면

$$v(0) = \frac{f}{b} + C = 0 \Rightarrow C = -\frac{f}{b}.$$

그러므로 $v(t)$의 특수식은 다음과 같이 쓸 수 있다.

$$v(t) = \frac{f}{b} - \frac{f}{b}e^{-(b/m)t} = \frac{f}{b}(1 - e^{-(b/m)t}). \tag{4}$$

(c)

$$\lim_{t\to\infty} v(t) = \lim_{t\to\infty} \frac{f}{b}(1 - e^{-(b/m)t}) = \frac{f}{b}.$$

6. (**총 10점, 각 5점**) Barrante 8장, 문제 3.

c

$$\mathbf{A}\langle 3, -1\rangle \cdot \mathbf{B}\langle 0, 4\rangle = 0 - 4 = -4$$

d

$$\mathbf{A}\langle 1, 1, 1\rangle \cdot \mathbf{B}\langle 2, 3, 4\rangle = 2 + 3 + 4 = 9$$

7. (**총 10점, 각 5점**) Barrante 8장, 문제 4.

(b)

$$\begin{aligned}
\mathbf{A}\langle -1, 2, 0\rangle \times \mathbf{B}\langle 2, 2, 0\rangle &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 2 & 0 \\ 2 & 2 & 0 \end{vmatrix}, \\
&= \mathbf{j}((2)\times(0) - (0)\times(2)) - \mathbf{j}((-1)\times(0) - (0)\times(2)) \\
&\quad + \mathbf{k}((-1)\times(2) - (2)\times(2)) \\
&= -6\mathbf{k}.
\end{aligned}$$

크기 $= \sqrt{(-6)^2} = 6.$

방향 $=$

$$\theta = \cos^{-1}\left(\frac{-6}{6}\right) = 180^\circ$$

$$\phi = \text{정할 수 없다.}$$

(c)

$$\begin{aligned}
\mathbf{A}\langle 3,-1,0\rangle \times \mathbf{B}\langle 0,4,0\rangle &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -1 & 0 \\ 0 & 4 & 0 \end{vmatrix}, \\
&= \mathbf{i}((-1)\times(0) - (0)\times(4)) - \mathbf{j}((3)\times(0) - (0)\times(0)) \\
&\quad +\mathbf{k}((3)\times(4) - (-1)\times(0)) \\
&= 12\mathbf{k}.
\end{aligned}$$

크기 $= \sqrt{(12)^2} = 12.$

방향 $=$

$$\theta = \cos^{-1}\left(\frac{12}{12}\right) = 0^\circ$$
$$\phi = \text{정할 수 없다.}$$

8. **(5점)** Barrante, 8장, 문제 9.

0이 아닌 두 벡터 $\mathbf{A}$와 $\mathbf{B}$의 내적이 0이라면, $\mathbf{A}\cdot\mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos\theta_{AB} = 0 \Rightarrow \cos\theta_{AB} = 0 \Rightarrow \theta_{AB} = 90^\circ$ 이기 때문에 $\mathbf{A}$와 $\mathbf{B}$는 서로 직교한다.

$$\begin{aligned}
\mathbf{A}\cdot\mathbf{B} &= \left(\frac{1}{2}\mathbf{q}_1 + \frac{1}{2}\mathbf{q}_2 + \frac{1}{2}\mathbf{q}_3 + \frac{1}{2}\mathbf{q}_4\right)\cdot\left(\frac{1}{2}\mathbf{q}_1 - \frac{1}{2}\mathbf{q}_2 + \frac{1}{2}\mathbf{q}_3 - \frac{1}{2}\mathbf{q}_4\right) \\
&= \frac{1}{4}\mathbf{q}_1\cdot\mathbf{q}_1 - \frac{1}{4}\mathbf{q}_1\cdot\mathbf{q}_2 + \frac{1}{4}\mathbf{q}_1\cdot\mathbf{q}_3 - \frac{1}{4}\mathbf{q}_1\cdot\mathbf{q}_4 + \frac{1}{4}\mathbf{q}_2\cdot\mathbf{q}_1 - \frac{1}{4}\mathbf{q}_2\cdot\mathbf{q}_2 + \frac{1}{4}\mathbf{q}_2\cdot\mathbf{q}_3 - \frac{1}{4}\mathbf{q}_2\cdot\mathbf{q}_4 \\
&\quad +\frac{1}{4}\mathbf{q}_3\cdot\mathbf{q}_1 - \frac{1}{4}\mathbf{q}_3\cdot\mathbf{q}_2 + \frac{1}{4}\mathbf{q}_3\cdot\mathbf{q}_3 - \frac{1}{4}\mathbf{q}_3\cdot\mathbf{q}_4 + \frac{1}{4}\mathbf{q}_4\cdot\mathbf{q}_1 - \frac{1}{4}\mathbf{q}_4\cdot\mathbf{q}_2 + \frac{1}{4}\mathbf{q}_4\cdot\mathbf{q}_3 - \frac{1}{4}\mathbf{q}_4\cdot\mathbf{q}_4 \\
&= \frac{1}{4} - \frac{1}{4} + \frac{1}{4} - \frac{1}{4} = 0,
\end{aligned}$$

위 식에서

$$\mathbf{q}_i\cdot\mathbf{q}_j = \delta_{ij}.$$

그러므로 $\mathbf{A}$와 $\mathbf{B}$는 서로 직교한다.

9. (**총 10점, 각 5점**) Barrante 9장, 문제 1.

(e)

$$\begin{vmatrix} x & 1 \\ 1 & x \end{vmatrix} x^2 - 1$$

(g)

$$\begin{vmatrix} 1 & 2 & 3 \\ 3 & 0 & 1 \\ -1 & 4 & 2 \end{vmatrix} = 1 \times \begin{vmatrix} 0 & 1 \\ 4 & 2 \end{vmatrix} - 2 \times \begin{vmatrix} 3 & 1 \\ -1 & 2 \end{vmatrix} + 3 \times \begin{vmatrix} 3 & 0 \\ -1 & 4 \end{vmatrix}$$

$$\begin{aligned} &= 1 \times \big(0 \times 2 - 1 \times 4\big) - 2 \times \big(3 \times 2 - 1 \times (-1)\big\} \\ &\quad +3\big(3 \times 4 - 0 \times (-1)\big) \\ &= -4 - 14 + 36 = 18 \end{aligned}$$

10. (**총 10점, 각 5점**) Barrante 9장, 문제 2.

(c)

$$\begin{vmatrix} 2x & 4 \\ 2 & x \end{vmatrix} = 6 \Rightarrow 2x^2 - 8 = 2 \Rightarrow x^2 = 5 \Rightarrow x = \pm\sqrt{5}.$$

(d)

$$\begin{aligned} \begin{vmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{vmatrix} &= 2 \\ &= x \begin{vmatrix} x & 1 \\ 1 & x \end{vmatrix} - 1 \times \begin{vmatrix} 1 & 1 \\ 1 & x \end{vmatrix} + 1 \times \begin{vmatrix} 1 & x \\ 1 & 1 \end{vmatrix} \\ &= x\big\{x \times x - 1 \times 1\big\} - 1 \times \big\{1 \times x - 1 \times 1\big\} \\ &\quad +1 \times \big\{1 \times 1 - x \times 1\big\} \\ &= x(x^2 - 1) - (x - 1) + (1 - x) = x^3 - 3x + 2 = 2 \\ &\Rightarrow x^3 - 3x = 0 \Rightarrow x = 0, \pm\sqrt{3}. \end{aligned}$$

11. (**5점**) Barrante 9장, 문제 6.

(b)

$$\begin{matrix} x + 2y + 3z = -5 \\ -x - 3y + z = -14 \\ 2x + y + z = 1 \end{matrix} \Rightarrow \begin{pmatrix} 1 & 2 & 3 \\ -1 & -3 & 1 \\ 2 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -5 \\ -14 \\ 1 \end{pmatrix}$$

$$x = \frac{\begin{vmatrix} -5 & 2 & 3 \\ -14 & -3 & 1 \\ 1 & 1 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 3 \\ -1 & -3 & 1 \\ 2 & 1 & 1 \end{vmatrix}} = \frac{17}{17} = 1.$$

$$y = \frac{\begin{vmatrix} 1 & -5 & 3 \\ -1 & -14 & 1 \\ 2 & 1 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 3 \\ -1 & -3 & 1 \\ 2 & 1 & 1 \end{vmatrix}} = \frac{51}{17} = 3.$$

$$z = \frac{\begin{vmatrix} 1 & 2 & -5 \\ -1 & -3 & -14 \\ 2 & 1 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 3 \\ -1 & -3 & 1 \\ 2 & 1 & 1 \end{vmatrix}} = \frac{-68}{17} = -4.$$

12. (**5 점**)

$$\begin{pmatrix} \alpha - E & \beta \\ \beta & \alpha - E \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = 0,$$

비자명한 해를 갖기 위해서

$$\begin{vmatrix} \alpha - E & \beta \\ \beta & \alpha - E \end{vmatrix} = 0$$

$$\beta^3 \begin{vmatrix} \frac{\alpha - E}{\beta} & 1 \\ 1 & \frac{\alpha - E}{\beta} \end{vmatrix} = \beta^2 \begin{vmatrix} x & 1 \\ 1 & x \end{vmatrix} = 0$$

윗 식에서 $x = (\alpha - E)/\beta$.

$$x^2 - 1 = (x+1)(x-1) = 0$$

(a) $x = -1$, 즉 $E = \alpha + \beta$ 일 때,

$$\begin{aligned} -\beta c_1 + \beta c_2 = 0 \quad &\Rightarrow \quad c_1 = c_2 \\ \beta c_1 - \beta c_2 = 0 \quad &\Rightarrow \quad c_1 = c_2 \end{aligned}$$

정규화조건: $c_1^2 + c_2^2 = 1$ 을 이용하면,

$$c_1 = \frac{1}{\sqrt{2}} \quad \text{and} \quad c_2 = \frac{1}{\sqrt{2}}.$$

(b) $x = 1$, 즉 $E = \alpha - \beta$ 일 때,

$$\begin{aligned} \beta c_1 + \beta c_2 = 0 \quad &\Rightarrow \quad c_1 = -c_2 \\ \beta c_1 + \beta c_2 = 0 \quad &\Rightarrow \quad c_1 = -c_2 \end{aligned} \tag{5}$$

정규화조건: $c_1^2 + c_2^2 = 1$ 을 이용하면,

$$c_1 = \frac{1}{\sqrt{2}} \quad \text{and} \quad c_2 = -\frac{1}{\sqrt{2}}.$$

13. (**총 10점, 각 5점**) Barrante 10장, 문제 1.

(d)

$$\hat{\mathbf{D}}_x(x^3 y) = \frac{\partial}{\partial x}(x^3 y) = 3x^2 y.$$

(e)

$$\hat{\mathbf{D}}_x^2\big(x^2 y^3\big) = \hat{\mathbf{D}}_x\big(2xy^3\big) = 2y^3$$

14. (**총 10점, 각 5점**) Barrante 10장, 문제 2.

(b) 세 함수의 집합 $\{f(x), g(x), h(x)\}$ 에 대하여

$$\begin{aligned} \hat{\mathbf{D}}_x \sum\{f(x), g(x), h(x)\} = \hat{\mathbf{D}}_x\big(f(x) + g(x) + h(x)\big) \quad &= \quad \frac{\partial}{\partial x}\Big(f(x) + g(x) + h(x)\Big) \\ &= \quad f'(x) + g'(x) + h'(x). \end{aligned}$$

$$\begin{aligned} \sum \hat{\mathbf{D}}_x\{f(x), g(x), h(x)\} = \sum \frac{\partial}{\partial x}\{f(x), g(x), h(x)\} \quad &= \quad \sum\{f'(x), g'(x), h'(x)\} \\ &= \quad f'(x) + g'(x) + h'(x). \end{aligned}$$

$\big[\hat{\mathbf{D}}_x, \sum\big] = \hat{\mathbf{D}}_x \sum - \sum \hat{\mathbf{D}}_x = 0$ 이므로, 연산자들은 교환가능하다.

(d) 세 함수의 집합 $\{f(x), g(x), h(x)\}$ 에 대하여

$$\sum \sqrt{\{f(x), g(x), h(x)\}} = \sum \{\sqrt{f(x)}, \sqrt{g(x)}, \sqrt{h(x)}\} = \sqrt{f(x)} + \sqrt{g(x)} + \sqrt{h(x)}.$$

$$\sqrt{\sum \{f(x), g(x), h(x)\}} = \sqrt{(f(x) + g(x) + h(x))} = \sqrt{f(x) + g(x) + h(x)}.$$

$\left[\sum, \sqrt{}\right] = \sum \sqrt{} - \sqrt{} \sum \neq 0$ 이므로, 연산자들은 교환가능하지 않다.

15. (**5 점**) Barrante 10 장, 문제 5.

$$\begin{aligned}
[\hat{p}_x, \hat{x}]\phi(x) &= \left[\frac{\hbar}{i}\frac{\partial}{\partial x}, x\right]\phi(x) \\
&= \frac{\hbar}{i}\frac{\partial}{\partial x}(x\phi(x)) - x\frac{\hbar}{i}\frac{\partial}{\partial x}\phi(x) \\
&= \frac{\hbar}{i}\phi(x) + \frac{\hbar}{i}x\frac{\partial}{\partial x}\phi(x) - x\frac{\hbar}{i}\frac{\partial}{\partial x}\phi(x) \\
&= \frac{\hbar}{i}\phi(x).
\end{aligned}$$

그러므로,

$$[\hat{p}_x, \hat{x}] = \frac{\hbar}{i} \neq 0.$$

16. (**5 점**) Barrante 10 장, 문제 9.

$$\begin{aligned}
\hat{\mathbf{O}}\phi = \hat{\mathbf{O}}\big(xe^{ax}\big) &= \left(\frac{\partial^2}{\partial x^2} - \frac{2a}{x}\right)\big(xe^{ax}\big) \\
&= \frac{\partial^2}{\partial x^2}\big(xe^{ax}\big) - \frac{2a}{x}\big(xe^{ax}\big) \\
&= \frac{\partial}{\partial x}\big(e^{ax} + axe^{ax}\big) - 2ae^{ax} \\
&= \big(ae^{ax} + ae^{ax} + a^2xe^{ax}\big) - 2ae^{ax} \\
&= a^2xe^{ax}
\end{aligned}$$

그러므로 고유값은 a^2 이다.

17. (**5 점**) Barrante 10 장, 문제 12.

$$\begin{aligned}
\hat{\mathbf{M}} = \frac{h}{2\pi i}(\mathbf{r} \times \nabla) &= \frac{h}{2\pi i}\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{vmatrix} \\
&= \mathbf{i}\frac{h}{2\pi i}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right) - \mathbf{j}\left(x\frac{\partial}{\partial z} - z\frac{\partial}{\partial x}\right) + \mathbf{k}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right) \\
&= \mathbf{i}\frac{h}{2\pi i}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right) + \mathbf{j}\frac{h}{2\pi i}\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right) + \mathbf{k}\frac{h}{2\pi i}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right) \\
&= \mathbf{i}\hat{M}_x + \mathbf{j}\hat{M}_y + \mathbf{k}\hat{M}_z
\end{aligned}$$

$$\hat{M}_x = \frac{h}{2\pi i}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)$$
$$\hat{M}_y = \frac{h}{2\pi i}\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right)$$
$$\hat{M}_z = \frac{h}{2\pi i}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)$$